Agent Registration and Discovery Protocol (ARDP)

IETF Side Meeting — AI Ecosystem & Discovery

Author(s): Roberto Pioli

Presenter: Hesham Moussa

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https://datatracker.ietf.org/doc/draft-pioli-agent-discovery/

Agenda

- **Overview of draft-pioli-agent-discovery (ARDP)**
  - Problem Statement: Finding Agents in a Dynamic World
  - Key Design Goals
  - Architecture and Components
  - Core Mechanisms
  - Security and Federation
- **ARDP's Relationship to Task Discovery (draft-mapmw-task-discovery)**
  - Complementary Framework
  - Incorporation of ARDP Concepts
  - Advantages and Standardization Opportunities
- Q&A

Problem Statement

- **Finding Agents in a Dynamic World is Hard**
- **Agents are not static websites.** They are:
  - Ephemeral (spin up/down as needed)
  - Mobile (can move between execution environments)
  - Heterogeneous (built by different vendors, using different protocols)
- **Challenges:**
  - How do you give an agent a **stable name** when its location changes?
  - How do clients find an agent's **current endpoint**?
  - How do you discover agents based on **what they can do** (their capabilities)?
  - How do you do this **securely**, preventing impersonation and unauthorized access?

ARDP solves this with stable identities (AIDs) + dynamic resolution.

ARDP: Agent Registration and Discovery Protocol

What is ARDP?

- ARDP is a lightweight, transport-agnostic control plane protocol for agent registration, discovery, and reachability in distributed environments
- Targets autonomous software agents that are Ephemeral, Mobile, and Heterogeneous
- Complements existing interaction protocols (e.g., MCP, A2A) by handling only discovery and registration.

Key challenges addressed:

- Stable addressing in dynamic environments
- Authorized, capability-based discovery
- Privacy-preserving presence signaling

ARDP Architecture and Operation in a glance

Core components:

- **Registrar:** accepts registrations, maintains AID ↔ endpoint bindings
- **Resolver:** answers authorized queries for AID → endpoints + capabilities

Agents register:

- AID
- Endpoints
- Capabilities document (JSON)
- TTL
- Proof-of-control (JWS)

Example transport: HTTPS + `/.well-known/ardp/` endpoints

Out of scope: agent-to-agent comms, full IAM, auditing

ARDP - Key Concepts (1)

Stable Identity for a Dynamic World

- **Agent Identifier (AID):** A stable, namespace identifier.
  - Syntax: agent:<local-id>@<authority>
  - Example: agent:customer-support@example.com
  - The <authority> denotes the administrative domain responsible for the identity. The <local-id> is stable, even if the agent's network location changes.

Dynamic Registration & Resolution

- **REGISTER:** Agents register their current endpoints (network location + protocol) and capabilities with an ARDP Registrar. This is "soft-state" and includes a Time-To-Live (TTL).
- **REFRESH:** Agents must periodically refresh their registration, making the system robust to agent churn and mobility.
- **RESOLVE:** Clients query a Resolver with an AID to get the current, active endpoint and capability set.
- **QUERY:** Authorized clients can search for agents by capability or namespace.

ARDP - Key Concepts (1)

Capability Advertisement:

- Agents declare their capabilities in a machine-readable document during registration.

- This includes:
  - **Supported Protocols:** MCP, A2A, HTTP, gRPC, etc.
  - **Protocol-specific Bindings:** Details needed to initiate an interaction.

- This allows a client, after discovering an agent, to select the right interaction protocol based on the task.

```
{
"protocols": ["A2A", "MCP"],
 "a2a_binding": { ... },
"mcp_binding": { ... }
}
```

Security

Built-in Security from the Ground Up

- Proof of Control (for Registration):
  - An agent must prove it controls the private key associated with its AID.
  - This is done via a JWS-signed payload that includes the registration request and a server-provided nonce, preventing replay attacks.
- Authorization Scopes (for Discovery):
  - Resolving an AID (RESOLVE) or performing a capability search (QUERY) requires specific authorization scopes (e.g., registry:resolve).
  - This ensures that only authorized clients can discover agent information.
- Privacy-Preserving Discovery:
  - QUERY responses are minimized by default, returning only basic status. Full details are available only with explicit request and authorization.

Federation and Scope

Designed for a Distributed World

- **Federation:**
  - Registrars from different domains (e.g., providerA.com and providerB.org) can explicitly federate.
  - Federated responses include provenance metadata (origin, signature), allowing clients to verify the source and trust the information.

- **Clear Scope & Layering:**
  - ARDP is intentionally narrow. It handles discovery, not execution, governance, or billing.
  - It composes cleanly with other layers: Identity governance, transport security (TLS/mTLS), and interaction protocols (MCP/A2A).

Key Differences: ARDP vs A2A vs AGNTCY (agntcy)

Aspect	ARDP (Pioli draft)	A2A (Agent-to-Agent)	AGNTCY (agntcy / Internet of Agents)
Primary Focus	Registration, stable AID, dynamic discovery & resolution only	Agent ↔ agent messaging, task delegation, coordination	Full stack: discovery + identity + secure messaging + observability
Scope	Minimal control plane (explicitly no comms/interaction)	Interaction & collaboration layer	Broad ecosystem infrastructure (superset/glue layer)
Discovery Mechanism	Registrar + Resolver (REGISTER/RESOLVE/QUERY ops, soft-state, JWS proofs)	Basic Agent Cards (.well-known/agent.json) — discovery often external	OASF schema + federated Agent Directories (supports A2A cards + MCP)
Communication	Out of scope — no messaging	Yes — core (JSON/HTTP, gRPC, stateful sessions)	Yes — via SLIM (secure low-latency) + ACP/AGP protocols
Identity & Security	Stable AIDs, proof-of-control, scoped auth, federation	Agent Cards with basic auth hints	Full Agent Identity service, verification, quantum-safe options

From ARDP to draft-mapmw-task-discovery

In draft-mapmw-task-discovery, **active task discovery** for agentic networks is introduced where:

- Task owners post structured “task cards”
- Agents proactively discover tasks, evaluate fit, submit proposals
- Supports coalition formation

Shifts focus from passive registry lookup → agent-initiated matching

ARDP is explicitly surveyed and referenced in this draft.

Summary of draft-mapmw-task-discovery

A Billboard for Tasks

- **Task Owner Layer:** Entities that submit tasks.
- **Task Owner Access Layer:** Interfaces for owners to interact with the platform.
- **Task-Posting Platform:** The billboard where tasks are published.
- **Agent Access Layer:** Interfaces for agents to discover tasks.
- **Agentic Layer:** Autonomous agents that evaluate and bid on tasks.
- **Communication Link:** Optional direct channel for negotiation.

How ARDP Relates to Task Discovery

- ARDP provides foundational **stable identity & secure registration**
- Used in task discovery for:
  - Validating agent identity in proposals
  - Resolving endpoints for communication
  - Verifying peers in coalitions
  - Establishing trust & provenance

Hybrid model: Agents register via ARDP → then actively discover tasks via mapmw framework

How ARDP Enables the task discovery Vision

Task discovery Requirement	ARDP Solution
Trustworthy Agent Identity	An agent applying for a task can present its AID. The task owner can use ARDP's RESOLVE to get the agent's authoritative endpoint and capabilities.
Coalition Formation	When agents team up, they need to find and trust each other. They can use ARDP to validate peer identities and resolve endpoints for coordination.
Authoritative Endpoint Resolution	The "Task Card" on the billboard doesn't need to duplicate agent metadata. It can simply reference an AID. The task owner or other agents can resolve the AID via ARDP to get current connection details.
Federated Trust	ARDP's federation model provides a way to verify agent provenance across domains, essential for a multi-provider task ecosystem.

IETF Standardization Opportunities

Potential IETF work:

- Standardize task card formats & APIs
- Define coalition primitives
- Integrate ARDP-style identity & proof mechanisms
- Add privacy, anti-gaming, audit features

Conclusion: ARDP

- A Lightweight Control Plane specifically designed for agent registration and discovery in distributed environments
- Stable Identity (AID) that decouples agent identity from dynamic network locations
- Soft-State Registration with TTL and refresh semantics, enabling agent mobility and ephemeral deployments
- Capability Advertisement that allows clients to discover not just where agents are, but what protocols they support (MCP, A2A, HTTP, gRPC)
- Security by Design with cryptographic proof-of-control for registration and fine-grained authorization scopes for discovery
- Federation Ready with explicit trust relationships and provenance metadata for cross-domain interoperability